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Time Series Model to Predict Wildfire Occurrence in California

1. Introduction

Recently, in Korea, there has been a huge wildfire incident that has been prolonged for almost two months. Before the incident, I thought wildfire was a minor incident that could be terminated within a day, so I was shocked when I saw that the news about the prolonged wildfire was all over the place. Due to the magnitude of the incident, I got personally curious about the factors that affect wildfires.

Causes for wildfires can be classified as human-driven or naturally occurring, but the former is the dominant reason. In California, one of the well-known regions for having frequent wildfires, about “ninety-five percent of wildfires in [the region] are caused by human drivers” such as making campfires, inappropriate handling of tobacco, or indirect causes for industrial activities (Chakraborty and Composto). The environmental conditions that make the area vulnerable to wildfire are high temperatures, strong wind, and low humidity, and global warming is worsening these conditions. The literature written by Farid et al. ran a simulation to “estimate the effect of climate change on rural fires in the Mediterranean region,” and it is reported that global warming, depending on their magnitude, would increase the amount of burned area “from 40% to 100%.” (Farid et al.)

Wildfire has many significant consequences that could have a negative impact on nature. The wildfire causes high emissions of carbon, and this change in the environment causes “biodiversity loss, global warming,” and other irreversible environmental changes (Singh). There are many other detrimental changes that wildfire can cause, and eventually, these environmental changes will become a consequence that humans must deal with. For this reason, I aim to construct a model that estimates wildfire occurrence and investigates the elements that affect wildfire and their weights. Because of the data availability, I have chosen to build a time series model on a dataset that was gathered by a research team at Georgia Southern University that has recorded the occurrence of wildfire in California from 1984 to 2025. From this research, I aim to narrow down the main focus of wildfire issues, so that measures to prevent wildfires can be established accordingly.

2. Model and Describe Your Approach

The equation and OLS estimation result from Table I show the time series model that has been created:

$$\text{Logit}(\text{FIRE_START_DAY}_t) = \beta_0 + \beta_1 \text{LAG_FIRE}_t + \beta_2 \text{MAX_TEMP}_t + \beta_3 \text{MIN_TEMP}_t + \beta_4 \text{AVG_WIND_SPEED}_t + \beta_5 \text{LAGGED_PRECIPITATION}_t + \beta_6 \text{LAGGED_AVG_WIND_SPEED}_t + \beta_7 \text{MONTH_1}_t + \beta_8 \text{MONTH_2}_t + \beta_9 \text{MONTH_3}_t + \beta_{10} \text{MONTH_4}_t + \beta_{11} \text{MONTH_5}_t + \beta_{12} \text{MONTH_6}_t + \beta_{13} \text{MONTH_7}_t + \beta_{14} \text{MONTH_8}_t + \beta_{15} \text{MONTH_9}_t + \beta_{16} \text{MONTH_10}_t + \beta_{17} \text{MONTH_11}_t + \beta_{18} \text{YEAR}_t + \varepsilon_i \quad (1)$$

Table I: Estimation Result of the Logistic Regression Model (1)

Number of Observation = 10,490

* p-value ≤ 0.05

Variable	Coefficient	Robust Standard Error	P-value
LAG FIRE _t	$\beta_1 = 0.6221404$	0.0620345	0.000 *
MAX TEMP _t	$\beta_2 = 0.0604353$	0.0054791	0.000 *

MIN_TEMP _t	$\beta_3 = 0.0023843$	0.0096929	0.806
AVG_WIND_SPEED _t	$\beta_4 = -0.0121907$	0.0212683	0.567
LAGGED_PRECIPITATION _t	$\beta_5 = -1.4884829$	0.2194607	0.000 *
LAGGED_AVG_WIND_SPEED _t	$\beta_6 = -0.0798866$	0.0354975	0.024 *
MONTH_1 _t	$\beta_7 = -0.3688041$	0.2183554	0.091
MONTH_2 _t	$\beta_8 = -0.5665746$	0.2511197	0.024 *
MONTH_3 _t	$\beta_9 = 0.0872886$	0.2125563	0.681
MONTH_4 _t	$\beta_{10} = 0.6218839$	0.2104213	0.003 *
MONTH_5 _t	$\beta_{11} = 1.5093198$	0.2056056	0.000 *
MONTH_6 _t	$\beta_{12} = 2.3065510$	0.2169207	0.000 *
MONTH_7 _t	$\beta_{13} = 2.5772431$	0.2324400	0.000 *
MONTH_8 _t	$\beta_{14} = 2.2934212$	0.2327200	0.000 *
MONTH_9 _t	$\beta_{15} = 1.6698201$	0.2223296	0.000 *
MONTH_10 _t	$\beta_{16} = 1.0167322$	0.1974862	0.000 *
MONTH_11 _t	$\beta_{17} = 0.4402444$	0.1887848	0.020 *
YEAR _t	$\beta_{18} = 0.0397471$	0.0035529	0.000 *
Constant	$\beta_0 = -85.3903029$	7.2439262	0.000 *

Data were preprocessed before the model creation for a smoother prediction. Null values were substituted with mean values, and dummy variables were created from time categorical variables such as month, year, and season to account for seasonality elements. To avoid multicollinearity problems, variables that could be represented as a linear combination of other variables were removed. For example, TEMP_RANGE was removed as it could be calculated from MAX_TEMP and MIN_TEMP, and WIND_TEMP_RATIO was removed as it could also be derived from AVG_WIND_SPEED and from the derived temperature value. At last, data were split into train and test sets in a seven-to-three ratio.

After adding all the environmental variables, different seasonal variables such as DAY_OF_YEAR, monthly dummy variables, and the year variable were added and deleted to test the best combination of the parameters. The train dataset was used to build the model, and the model was tested on the test dataset to create a prediction of a value between 0 and 1. Using 0.5 as a midpoint, the values that were bigger than or equal to 0.5 were classified as 1, and others were classified as 0. Based on this classification, accuracy was tested on both train and test datasets, and the model with the highest test dataset accuracy was chosen as the best model. The following Table II shows the accuracy values for different trials.

	Seasonal Variables	Train Dataset Accuracy	Test Dataset Accuracy
Trial 1	DAY_OF_YEAR	0.7771	0.7518
Trial 2	no seasonal variables	0.7769	0.7536
Trial 3	seasonal dummy variables	0.7842	0.7485
Trial 4	monthly dummy variables	0.7917	0.7547
Trial 8	monthly dummy variables + YEAR	0.7926	0.7569 (highest)

By testing different combinations of seasonal variables, the model from Trial 8 had the best test dataset accuracy. Different interaction terms were tried on that model, but the test dataset accuracy was not improved any further, so no further refinements were made from Trial 8.

3. Empirical Analysis

The original data *California Weather and Fire Prediction Dataset (1984-2025) with Engineered Features* was gathered by Yavas et al. It contains records of weather and time features. The following table shows the descriptions of the original features stated by the research team:

Table III: Description of Dataset Features	
Feature Name	Description
DATE	The date of the observation.
PRECIPITATION	Daily precipitation in inches.
MAX_TEMP	Maximum daily temperature in degrees Fahrenheit.
MIN_TEMP	Minimum daily temperature in degrees Fahrenheit.
AVG_WIND_SPEED	Average daily wind speed in mph.
FIRE_START_DAY	A binary indicator (True/False) showing whether a wildfire started on that date.
YEAR	The year of the observation.
TEMP_RANGE	The difference between maximum and minimum temperatures, indicating daily temperature variability.
WIND_TEMP_RATIO	The ratio of average wind speed to maximum temperature, capturing wind-temperature dynamics.
MONTH	The calendar month of the observation (1–12).
SEASON	The season of the observation (Winter, Spring, Summer, Fall).
LAGGED_PRECIPITATION	Cumulative precipitation over the preceding 7 days, reflecting recent moisture conditions.
LAGGED_AVG_WIND_SPEED	Average wind speed over the preceding 7 days, indicating sustained wind patterns.
DAY_OF_YEAR	The numeric day within the year (1–365/366).

The following Figure I shows the descriptive statistics of the variables.

DATE	PRECIPITATION	MAX_TEMP	MIN_TEMP	AVG_WIND_SPEED	FIRE_START_DAY
Length:14988	Min. :0.00000	Min. : 50.00	Min. :33.00	Min. : 1.790	Length:14988
Class :character	1st Qu.:0.00000	1st Qu.: 65.00	1st Qu.:51.00	1st Qu.: 6.040	Class :character
Mode :character	Median :0.00000	Median : 70.00	Median :57.00	Median : 7.160	Mode :character
	Mean :0.03232	Mean : 70.53	Mean :56.49	Mean : 7.435	
	3rd Qu.:0.00000	3rd Qu.: 75.00	3rd Qu.:62.00	3rd Qu.: 8.500	
	Max. :4.53000	Max. :106.00	Max. :77.00	Max. :26.170	
	NA's :1	NA's :1	NA's :1	NA's :12	
YEAR	TEMP_RANGE	WIND_TEMP_RATIO	MONTH	SEASON	LAGGED_PRECIPITATION
Min. :1984	Min. : 2.00	Min. :0.02355	Min. : 1.000	Length:14988	Min. :0.0000
1st Qu.:1994	1st Qu.:10.00	1st Qu.:0.08524	1st Qu.: 4.000	Class :character	1st Qu.:0.0000
Median :2004	Median :12.00	Median :0.10222	Median : 7.000	Mode :character	Median :0.0000
Mean :2004	Mean :14.04	Mean :0.10702	Mean : 6.518		Mean :0.2262
3rd Qu.:2014	3rd Qu.:17.00	3rd Qu.:0.12046	3rd Qu.:10.000		3rd Qu.:0.0600
Max. :2025	Max. :41.00	Max. :0.45912	Max. :12.000		Max. :8.1800
	NA's :1	NA's :12			
LAGGED_AVG_WIND_SPEED	DAY_OF_YEAR				
Min. : 3.227	Min. : 1				
1st Qu.: 6.519	1st Qu.: 92				
Median : 7.479	Median :183				
Mean : 7.434	Mean :183				
3rd Qu.: 8.279	3rd Qu.:274				
Max. :13.933	Max. :366				

Figure I. Descriptive Statistics of all the Variables in the Original Dataset

The model that is predicted can be revisited by looking at Table I. To investigate the seasonal components that are included in the equation, graphs have been created to record mean occurrence of wildfire by two different time divisions – monthly and yearly divisions.

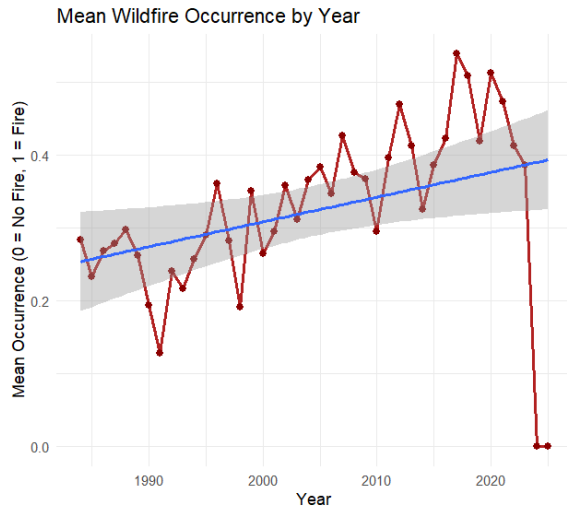


Figure II. Graph of the Mean Wildfire Occurrence by Year

The literature that was mentioned in the introduction states that global warming has been raising the chances of wildfire in the past. The visualization in Figure II supports that there has been an annual trend to raise the chances of wildfire. An oddity is that there have been zero chances of wildfires for the years 2024 and 2025. It is understandable that the year 2025 is reported to have zero incidents of wildfires, as the data was only recorded until January of 2025, but it does not make sense that the year 2024 was recorded to have zero wildfires. Any incidents of wildfire from 2024 could be easily searched for on the internet, so there seems to be an imprecise side to the data.

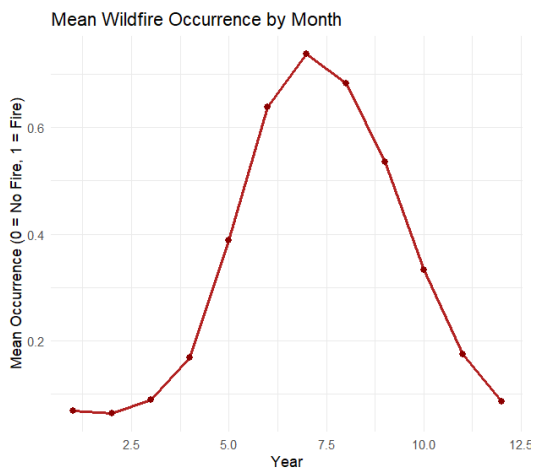


Figure III. Graph of the Mean Wildfire Occurrence by Month

The visualization in Figure III captures the monthly trend of fire occurrence. The months that correspond to the summer season are shown to have a higher occurrence of wildfire. This supports the literature's assertion that wildfire is prone in high-temperature environments.

In Table II, the best model had a train set accuracy of 0.7926 and a test set accuracy of 0.7569. When the train set accuracy is higher than that of the test set, the model might be prone to overfitting. In our case, the difference is less than 0.05, so overfitting isn't a big thing to worry about. 0.5 was used as a point to make binary classifications, but this can be adjusted depending on whether getting fewer false positives or false negatives is important. In Table I, most of the estimators had small p-values except for

the following features: MIN_TEMP, AVG_WIND_SPEED, MONTH_3. These values may seem unnecessary, but they are features that theoretically increase the possibility of wildfire, so including them seemed necessary. Additionally, removing them lowered the prediction accuracy of the train dataset, so to get the optimal model, these variables were kept.

4. Robustness Check

As mentioned in the Empirical Analysis, the record for FIRE_START_DAY in 2024 is suspicious. According to the *2024 Incident Archive* by California Department of Forestry and Fire Protection, there have been 8,110 wildfires recorded in California. There might be different standards of whether to consider an incident a wildfire or not, but the gap between zero incidents and 8,110 incidents seems so huge to be attributed to different classification standards. Data in 2024 are all included in the test dataset, so this possible error of classification would have lowered our accuracy for the test dataset. Thus, there's a possibility that the model would have a better performance with more precise data.

5. Conclusion

Overall, the model incorporates environmental features that are major predictors of wildfires – for instance, MAX_TEMP, AVG_WIND_SPEED, and LAGGED_PRECIPITATION. For a more precise prediction, monthly dummy variables and YEAR variable are added to account for seasonal trends, and we were able to come up with a final model that had an accuracy of 0.7569.

With environmental data collected in the future, the developed model can be used to calculate the possibility of wildfire occurrence. The prediction can be used to indicate periods of time that are vulnerable to wildfires, and prevention measures can be taken accordingly. The model is limited in that it cannot indicate a specific location of the wildfires inside California, so future research would be needed to locate the dangerous area with the environmental variables we have.

6. Works Cited

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